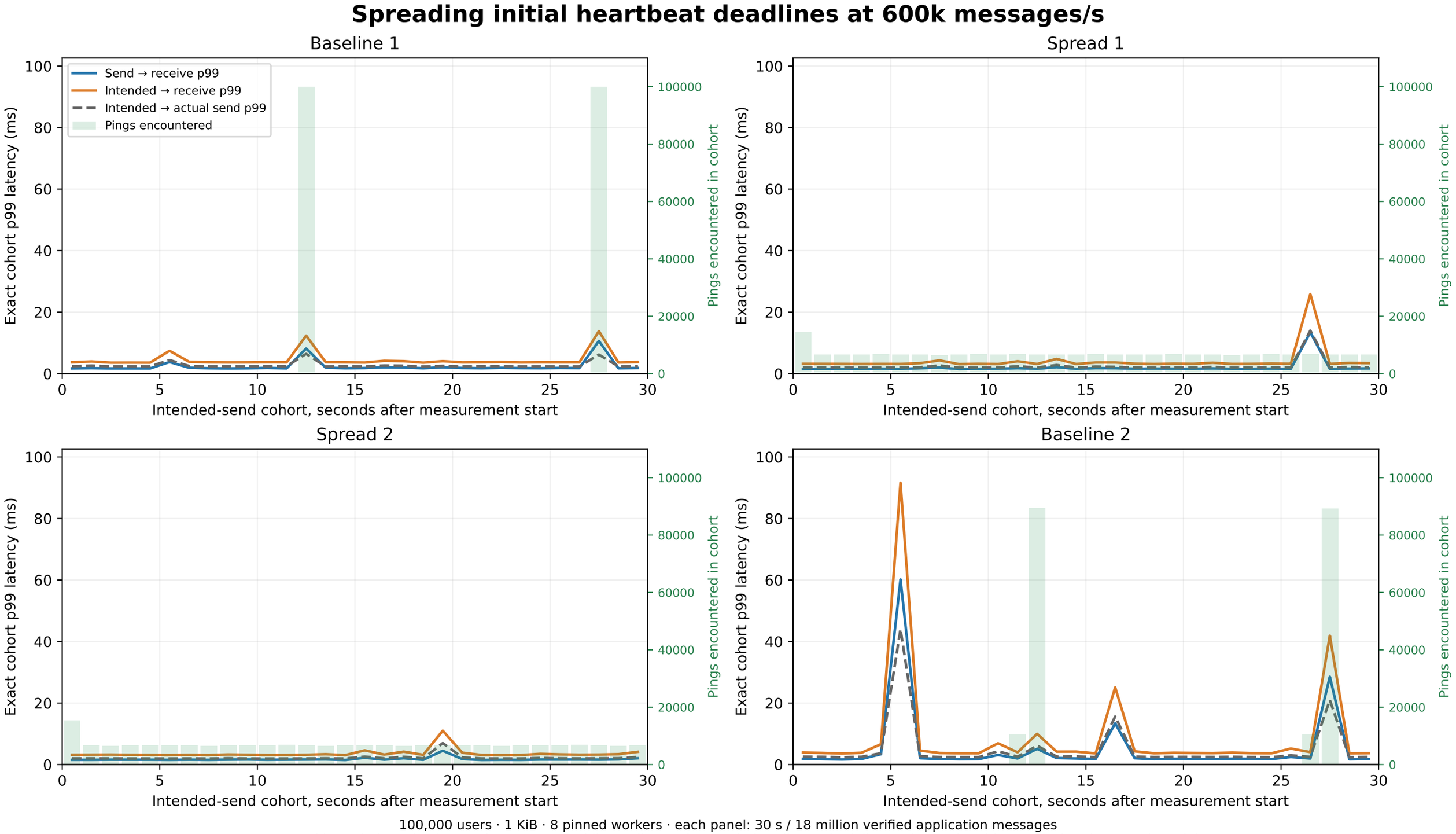
Spreading initial heartbeat deadlines at 600k messages/s
Baseline 1
Spread 1
Spread 2
Baseline 2
Send → receive p99
Intended → receive p99
Intended → actual send p99
Pings encountered
Exact cohort p99 latency (ms)
Pings encountered in cohort
Intended-send cohort, seconds after measurement start
100,000 users · 1 KiB · 8 pinned workers · each panel: 30 s / 18 million verified application messages
One-second intended-send cohorts; Pings counted while reading probes. Includes client scheduling and room batching; no isolated server-time claim.